

1. The matrix corresponding to the multiplication of $X^2 + 3X + 2$ modulo $X^3 + 1$ is:

$$= \begin{pmatrix} 2 & -1 & -3 \\ 3 & 2 & -1 \\ 1 & 3 & 2 \end{pmatrix}$$

2. Maple Worksheet Attached.
3. Maple Worksheet Attached.